

LAB NOC 03

Cisco Catalyst 2960 — EtherChannel Failure / VLAN Mismatch Troubleshooting

Real Hardware | ES / EN | NOC Portfolio

INFORMACION DEL LABORATORIO / LAB INFORMATION

Hardware / Hardware	Cisco Catalyst 2960 (WS-C2960-24TT-L)	Cisco Catalyst 2960 (WS-C2960-24TT-L)
Categoría / Category	Troubleshooting EtherChannel / VLAN	EtherChannel / VLAN Troubleshooting
Nivel / Level	CCNA — Switching / Troubleshooting	CCNA — Switching / Troubleshooting
Duración / Duration	Aprox. 60 minutos	Approx. 60 minutes
Entorno / Environment	Hardware real (equipo usado)	Real hardware (used equipment)
Tecnologías	VLAN, EtherChannel, Forced mode, Troubleshooting	VLAN, EtherChannel, Forced mode, Troubleshooting
Relevancia NOC	Resolución de incidente L2, cierre de ticket	L2 incident resolution, ticket closure

OBJETIVO / OBJECTIVE

OBJETIVO	OBJECTIVE
Diagnosticar y resolver una falla de EtherChannel causada por VLAN mismatch en un Cisco Catalyst 2960 real. Documentar el incidente en formato NOC (Incident Report), ejecutar los pasos de recuperación y verificar el restablecimiento del servicio.	Diagnose and resolve an EtherChannel failure caused by VLAN mismatch on a real Cisco Catalyst 2960. Document the incident in NOC format (Incident Report), execute recovery steps, and verify service restoration.

PROCEDIMIENTO — 4 ETAPAS / PROCEDURE — 4 STAGES

#	ESPAÑOL	ENGLISH
1	ETAPA 1 — Baseline: verificar estado inicial del switch	STAGE 1 — Baseline: verify initial switch state
2	Ejecutar: show vlan brief → confirmar VLAN 1 activa, VLANs 1002-1005 reservadas	Run: show vlan brief → confirm VLAN 1 active, VLANs 1002-1005 reserved
3	Ejecutar: show etherchannel summary → detectar Port-Channel1 (Po1) preexistente del lab anterior	Run: show etherchannel summary → detect pre-existing Port-Channel1 (Po1) from previous lab
4	Ejecutar: show interfaces status → Fa0/6 suspended VLAN 10, Fa0/16 suspended VLAN	Run: show interfaces status → Fa0/6 suspended VLAN 10, Fa0/16 suspended VLAN

	20, Po1 DOWN	20, Po1 DOWN
5	ETAPA 2 — Diagnostico: identificar causa raiz	STAGE 2 — Diagnosis: identify root cause
6	Identificar error en syslog: EC-5-CANNOT_BUNDLE2 — Fa0/6 not compatible with Po1	Identify syslog error: EC-5-CANNOT_BUNDLE2 — Fa0/6 not compatible with Po1
7	Confirmar: Fa0/6 en VLAN 10 y Fa0/16 en VLAN 20 — VLAN mismatch entre miembros del bundle	Confirm: Fa0/6 on VLAN 10 and Fa0/16 on VLAN 20 — VLAN mismatch between bundle members
8	Causa raiz verificada: EtherChannel requiere configuracion identica en todos sus puertos miembro	Root cause verified: EtherChannel requires identical configuration on all member ports
9	ETAPA 3 — Resolucion: recuperar el servicio	STAGE 3 — Resolution: restore service
10	default interface fa0/6 y fa0/16 — resetear puertos a configuracion de fabrica	default interface fa0/6 and fa0/16 — reset ports to factory configuration
11	no interface port-channel 1 — eliminar Po1 (puertos pasan a administratively down)	no interface port-channel 1 — delete Po1 (ports go administratively down)
12	Reconfigurar Fa0/6 y Fa0/16: mode access, VLAN 10 en ambos, channel-group 1 mode on	Reconfigure Fa0/6 and Fa0/16: mode access, VLAN 10 on both, channel-group 1 mode on
13	ETAPA 4 — Verificacion: show etherchannel summary → Po1 estado U (in use), puertos activos	STAGE 4 — Verification: show etherchannel summary → Po1 state U (in use), ports active
14	Confirmar en syslog: LINEPROTO-5-UPDOWN: Interface Fa0/6 changed state to up (idem Fa0/16)	Confirm in syslog: LINEPROTO-5-UPDOWN: Interface Fa0/6 changed state to up (same Fa0/16)

INCIDENT REPORT — FORMATO NOC / NOC FORMAT

CAMPO / FIELD	DETALLE / DETAIL
Device / Equipo	Cisco Catalyst 2960 (WS-C2960-24TT-L)
Ports Affected / Puertos afectados	Fa0/6, Fa0/16
Service Affected / Servicio afectado	Layer 2 connectivity — EtherChannel bundle Po1
Symptoms / Sintomas	Puertos muestran estado 'suspended' / Ports show 'suspended' status
Root Cause / Causa raiz	VLAN mismatch entre puertos miembros del EtherChannel / VLAN mismatch between EtherChannel member ports
Error Message	EC-5-CANNOT_BUNDLE2: Fa0/6 is not compatible with Po1
Resolution	default interface fa0/6 + fa0/16 → no interface port-channel 1 → reconfigure with matching VLAN → channel-group 1 mode on

REFERENCIA DE COMANDOS / COMMAND REFERENCE

COMANDO / COMMAND	SIGNIFICADO (ES)	MEANING (EN)
<code>show vlan brief</code>	Verifica VLANs activas y asignacion de puertos	Verifies active VLANs and port assignments
<code>show interfaces status</code>	Muestra estado de cada puerto (connected/suspended/disabled)	Shows each port status (connected/suspended/disabled)
<code>show etherchannel summary</code>	Muestra estado del bundle EtherChannel y sus miembros	Shows EtherChannel bundle status and member ports
<code>default interface fa0/X</code>	Resetea el puerto a configuracion de fabrica	Resets the port to factory configuration
<code>no interface port-channel 1</code>	Elimina el Port-Channel y libera los puertos miembro	Deletes the Port-Channel and releases member ports
<code>channel-group 1 mode on</code>	Fuerza la formacion del EtherChannel (sin LACP/PAgP)	Forces EtherChannel formation (without LACP/PAgP)
<code>switchport mode access</code>	Configura el puerto en modo acceso	Sets port to access mode
<code>switchport access vlan 10</code>	Asigna el puerto a VLAN 10	Assigns port to VLAN 10

RESOLUCION PASO A PASO / STEP-BY-STEP RESOLUTION

#	COMANDO / COMMAND	ACCION (ES)	ACTION (EN)
1	<code>default interface fastEthernet 0/6</code>	Resetea Fa0/6 a fabrica (limpia EtherChannel, VLAN, mode)	Resets Fa0/6 to factory (clears EtherChannel, VLAN, mode)
2	<code>default interface fastEthernet 0/16</code>	Resetea Fa0/16 a fabrica	Resets Fa0/16 to factory
3	<code>no interface port-channel 1</code>	Elimina Po1 — puertos quedan administratively down	Deletes Po1 — ports go administratively down
4	<code>interface fa0/6 → switchport mode access</code>	Configura modo acceso en Fa0/6	Configures access mode on Fa0/6
5	<code>switchport access vlan 10</code>	Asigna VLAN 10 a Fa0/6 (igual que Fa0/16)	Assigns VLAN 10 to Fa0/6 (matching Fa0/16)
6	<code>channel-group 1 mode on</code>	Re-agrega Fa0/6 al bundle forzado	Re-adds Fa0/6 to forced bundle
7	<code>interface fa0/16 → misma configuracion</code>	Replica configuracion identica en Fa0/16	Replicates identical config on Fa0/16
8	<code>show etherchannel summary</code>	Verifica bundle Po1 en estado U (in use)	Verifies bundle Po1 in state U (in use)

TROUBLESHOOTING DOCUMENTADO / DOCUMENTED TROUBLESHOOTING

HALLAZGO / FINDING	ACCION CORRECTIVA / CORRECTIVE ACTION
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Po1 DOWN y puertos en estado 'suspended' al iniciar el lab. / Po1 DOWN and ports in 'suspended' state at lab start.	Identificado como configuracion residual del lab anterior (EtherChannel). No afecto el lab actual pero requirio diagnostico antes de proceder. / Identified as residual config from previous lab. Did not affect current lab but required diagnosis before proceeding.
Error EC-5-CANNOT_BUNDLE2 persistio despues de hacer default interface en los puertos. / EC-5-CANNOT_BUNDLE2 persisted after running default interface on ports.	La causa fue que Po1 aun tenia configuracion vieja. Solucion: eliminar el Po1 con no interface port-channel 1 antes de reconfigurar los puertos miembro. / Cause: Po1 still had old config. Solution: delete Po1 with no interface port-channel 1 before reconfiguring member ports.
Tras no interface port-channel 1, los puertos quedaron administratively down. / After no interface port-channel 1, ports went administratively down.	Comportamiento esperado en Cisco IOS. Se reconfiguraron los puertos con mode access + VLAN 10 + channel-group 1 mode on. Puertos volvieron a estado up. / Expected Cisco IOS behavior. Ports reconfigured with mode access + VLAN 10 + channel-group 1 mode on. Ports returned to up state.
En laboratorio se documento la falla sin restaurar servicio en primera instancia (metodologia valida en training). / In lab, failure was documented without restoring service initially (valid training methodology).	Se documento el estado de falla (Incident Report) y luego se ejecuto la resolucion completa. Ambas opciones son validas en NOC: restaurar o documentar segun contexto. / Failure state documented (Incident Report), then full resolution executed. Both options valid in NOC: restore or document depending on context.

RESULTADO / RESULT

RESULTADO (ES) Falla de EtherChannel diagnosticada (VLAN mismatch), documentada en Incident Report formato NOC, y resuelta con exito. Po1 restaurado a estado activo. Syslog confirma interfaces up. Ticket cerrado.	RESULT (EN) EtherChannel failure diagnosed (VLAN mismatch), documented in NOC-format Incident Report, and successfully resolved. Po1 restored to active state. Syslog confirms interfaces up. Ticket closed.
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KEY TAKEAWAY

"En un EtherChannel todos los puertos miembro deben tener configuracion identica: mismo mode, misma VLAN, mismo tipo de negociacion. Un solo mismatch suspende todo el bundle. En NOC el proceso es: baseline → diagnostico → Incident Report → resolucion → verificacion → cierre de ticket. No se toca un puerto en produccion sin documentar primero el estado de falla."

"In an EtherChannel all member ports must have identical configuration: same mode, same VLAN, same negotiation type. A single mismatch suspends the entire bundle. In NOC the process is: baseline → diagnosis → Incident Report → resolution → verification → ticket closure. No production port is touched without first documenting the failure state."

NOC RELEVANCE — MINING / INDUSTRIAL ENVIRONMENT

In a mining operation (e.g., Newmont Cerro Negro), EtherChannel links are used to aggregate bandwidth between core and distribution switches — carrying SCADA traffic, IP cameras, and VoIP

simultaneously. A Po going down due to VLAN mismatch can silently disconnect multiple systems. A NOC engineer must: (1) quickly identify suspended ports via show interfaces status, (2) correlate with EC-5-CANNOT_BUNDLE2 in syslog, (3) determine if the mismatch is a config error or an unauthorized change, (4) open an Incident Report before touching anything in production, and (5) restore service following a tested procedure — exactly as documented in this lab.